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Title: WPT - Transmitter Block

Size: A3	Date: 2020-02-22	**Rev:**
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12–24V DC Input

J201
54–00129

TP208 TP1.0
TP209 TEST–1
GND

24V full–scale
ADC_VDC = 24 * 3.6k / (3.6k + 75k) = 1.099 V

R201 75kF
TP203 ADC_VDC
TP1.0
R202 3.6kF
GND

U201
TS30042–M000QFNR

C205 0.022u/50(X7R)
L201 NR5040T4R7N
FB201 BLM18PG330SN1D

VCC(abs.max)=42V
BST VSW VSW VSW
VCC VCC VCC
EN/SYNC FB
PG PAD PGND PGND
NC NC GND

C201 10u/50(X7R)
C203 0.1u/50(X7R)
C207 0.1u/50(X7R)
C212 10u/50(X7R)
C218 10u/50(X7R)
C220 10u/50(X7R)
C222 10u/50(X7R)
C223 10u/50(X7R)

D202 S210FA

TP219 TP1.0
R206 10kF
R211 100F
TP221 TP1.0
TP222 TP1.0
R210 3.9kF

R212 10kF
PWRGD_HV
TP210 TEST–1

R213 0
HVDCDC_EN
R207 10kF
D201 OSR5JA3Z74A

Ext. DC Input < 40 V
Select by Jumber

CN203
ULO–PH409–1*3SF11

Vout
= 24 V(@Rab: 100k ohm)
= 0.9V(@Rab: 0 ohm)

R205 27mF(1/4W)
PWR_FLAG
VPP
TP206 TEST–1

Through Holes for Extention

Arduino Nano Connector

CN201 ULO–FH411–1*15SF11
D1(TX) D0(RX) RESET GND D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12
HVDCDC_EN
TP207 TP1.0
TP204 TP1.0 WLAT
TP205 TP1.0 ALERT

CN202 ULO–FH411–1*15SF11
CN204 ULO–FH411–1*15SF11
VIN GND RESET 5V A7 A6 SCL/A5 SDA/A4 A3 A2 A1 A0 AREF 3.3V D13
TP201 TP1.0 SCL
TP202 TP1.0 SDA
ADC_VDC

CN205 ULO–FH411–1*15SF11

AREF require 1.1V internal reference

U204
MCP45HV51–104E_ST

C209 0.1u/50(X7R)
V+ to V–: 40V(abs.max)
VL DGND V+ V– POA POW POB SCL SDA A1 A0 WLAT SHDN NC
SCL_5V
SDA_5V
R223 0
WLAT
R225 10kF

Full Scale: 100k ohm
256 tap
I2C Address(7–bit)
0b0111100

U203
PAC1720–1–AIA–TR
SENSE1+ SENSE1– SENSE2+ SENSE2– VDD ADDR_SEL SMCLK SMDATA GND EP ALERT
C211 0.1u/50(X7R)
SCL_5V
SDA_5V
ALERT
R224 0
I2C Address(7–bit)
0b1001100

U202
PCA9306DCUR
GND Voltage I2C1 < I2C2 EN VREF1 VREF2 SCL1 SCL2 SDA1 SDA2
R214 200kF
R215 10kF
R216 10kF
R217 10kF
R218 10kF
R219 OPEN
R220 OPEN
C210 0.1u/50(X7R)
SCL_3.3V
SDA_3.3V
SCL
SDA
R221 0
R222 0

CN206
SM04B–SRSS–TB(LF)(SN)
R203 0
R204 OPEN
Connect to
Receiver
(For Debug)
SDA
SCL
FG

CN207
ULO–PH409–1*3SF11
Select by Jumber
TP211 TP1.0

J202
ULO–PH409–1*2SF11
Throught Hole
for External DC Input
(24–40 V)
TP216 TP1.0
TP217 TEST–1

U207
TS30042–M000QFNR

C206 0.022u/50(X7R)
L202 NR5040T4R7N
FB202 BLM18PG330SN1D

C202 10u/50(X7R)
C204 0.1u/50(X7R)
C208 0.1u/50(X7R)
C217 22u/16(X5R)
C219 22u/16(X5R)
C221 10n/50(X7R)

D203 S210FA

TP220 TEST–1
R208 68kF
R209 10kF
+7V

R234 10kF
PWRGD_5V
TP218 TEST–1

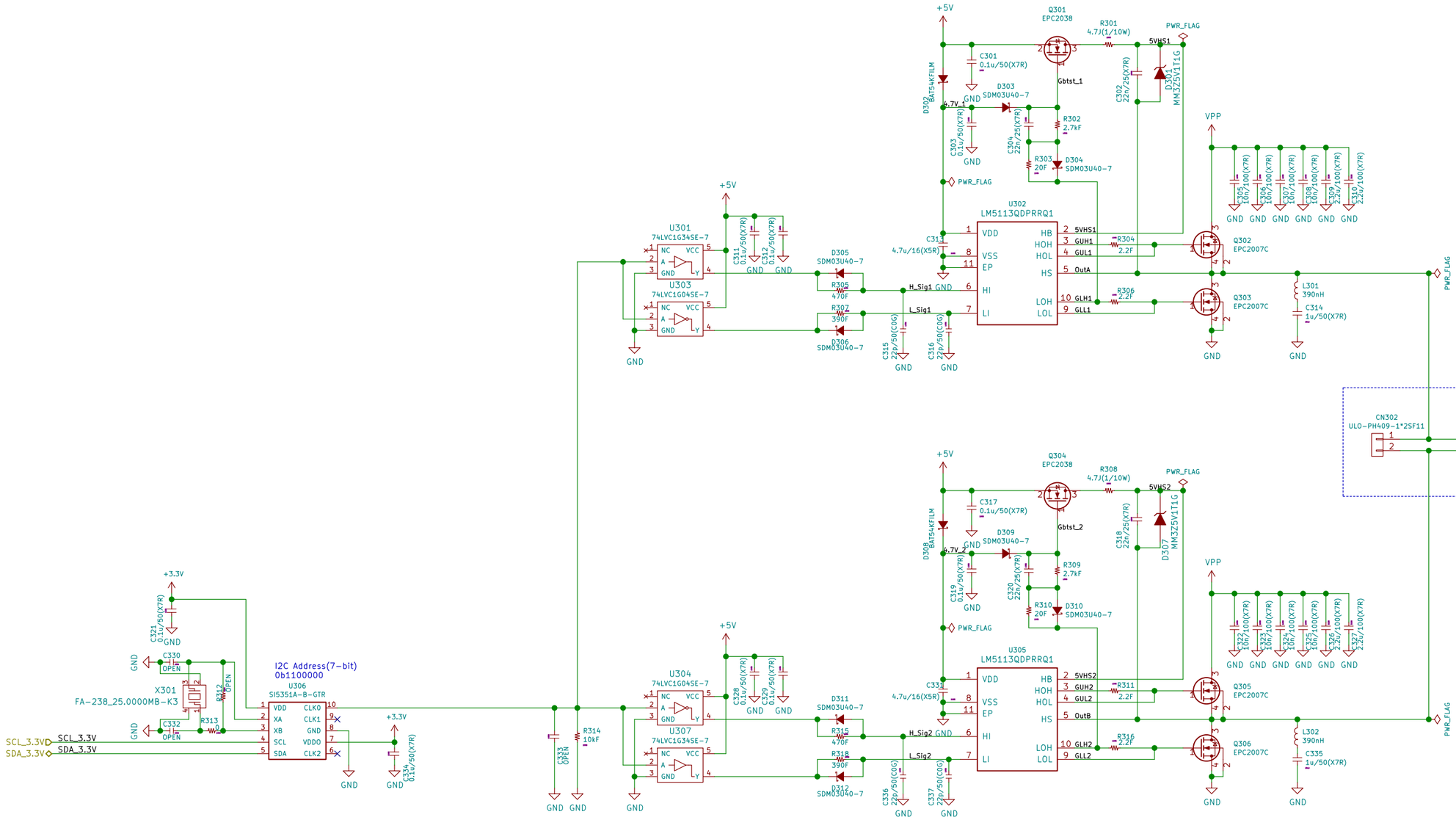
VIN: Power Supply for Arduino
VIN

U205
NCP1117LPST50T3G
LDO: 5V/1A
VI VO GND
C215 10u/25(X5R)
C216 10u/25(X5R)
R226 27mF(1/4W)
TP212 TEST–1

U206
MIC5504–3.3YM5–TR
VI VO GND EN NC
C213 1u/50(X7R)
C214 1u/50(X7R)
TP223 TEST–1
TP224 TEST–1
TP214 TEST–1
TP215 TEST–1

+5V/GND Through Holes for External Devices
TP213 TEST–1
TP225 TEST–1
CN208 ULO–PH409–1*2SF11
CN209 ULO–PH409–1*2SF11
CN210 ULO–PH409–1*2SF11

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